# Solution Requirements

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| **Date** | 15 March 2026 |
| **Team ID** | SWTID-2026-2085 |
| **Project Name** | OptiCrop: Smart Agricultural Production Optimization Engine |
| **Maximum Marks** | 4 Marks |

**Define Functional and Non-Functional Requirements:**

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

**Step 1: Functional Requirements (FR)**

| S.No | Requirement Category | Requirement Description | Priority (High/Medium/Low) |
|---|---|---|---|
| 1 | Authentication | Farmers can securely access the system using a username and password (or access the application without login if authentication is not implemented). | High |
| 2 | Authorization levels | Users can enter soil and environmental data and view crop recommendations. Administrators can manage datasets and the machine learning model. | High |

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| 3 | External interfaces | The system accepts soil parameters (N, P, K, pH) and environmental data (temperature, humidity, rainfall). It may integrate with weather APIs in future versions. | Medium |
| 4 | Transactions processing | The system validates user input, processes it through the machine learning model, and returns the most suitable crop recommendation. | High |
| 5 | Reporting | Display crop prediction results along with relevant crop information. Future versions may include downloadable reports. | Medium |
| 6 | Business rules, etc. | Crop recommendations are generated only after validating all required soil and environmental parameters. Predictions are based on the trained machine learning model. | High |
| 7 | Compliance to laws or regulations | User data should be handled securely and follow applicable data privacy guidelines. | Medium |
| 8 | *Other* | The system should provide accurate crop recommendations based on the trained dataset and allow easy interaction through a user-friendly interface. | High |

**Step 2: Non-Functional Requirements (NFR)**

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|---|---|---|---|
| **1** | Performance & Speed | The system should generate crop recommendations quickly after receiving input. | Prediction generated in **less than 3 seconds**. |
| **2** | Scalability | The system should support multiple farmers using the application simultaneously and allow future dataset expansion. | Supports **100+ concurrent users**. |
| **3** | Security & Data Privacy | User inputs and prediction data should be protected from unauthorized access. | Secure data handling and authenticated access (if implemented). |
| **4** | Reliability & Availability | The application should remain available with minimal downtime. | **99% uptime** during normal operation. |
| **5** | Usability & Accessibility | The interface should be simple, intuitive, and easy for farmers to use on desktop or mobile devices | Users can complete a prediction in **3 simple steps**. |
| **6** | *Other* | The system should be easy to update with new datasets or machine learning models and deploy on different platforms. | Modular design with easy maintenance and deployment. |